

A Comparative Analysis of Four Commonly Implemented Declustering Algorithms

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Abstract

Declustering of earthquake catalogs, that is determining dependent and independent events in an earthquake sequence, is a common feature of many seismological studies. While many different declustering algorithms exist, each has different performance and sensitivity characteristics. Here, we conduct a comparative analysis of the four most commonly used declustering algorithms: Gardner and Knopoff (1975), Reasenberg (1985), Zhuang et al. (2002), and Zaliapin and Ben-Zion (2008) in four different tectonic settings. Overall, we find that the Zaliapin and Ben-Zion (2008) algorithm effectively removes aftershock sequences, while simultaneously retaining the most information (i.e. the most events) in the output catalog and not significantly modifying statistical characteristics (i.e. the Gutenberg Richter b-value). Both Gardner and Knopoff (1975) and Zhuang et al. (2002) also effectively remove aftershock sequences, though they remove significantly more events than the other algorithms. By contrast, Reasenberg (1985) only effectively removed aftershocks in one of the test regions.

Main Points

- We test the performance of four common declustering algorithms on four different source catalogs from four different tectonic settings.
- We compare the algorithms in terms of statistical variations between source and output catalog and data loss.
- We find the nearest neighbor approach of Zaliapin and Ben-Zion (2008) works best in declustering any given earthquake catalog.

1. Introduction

A feature ubiquitous among earthquake catalogs is the presence of a large number of dependent events, that is foreshocks and aftershocks, clustered in space and time around a mainshock. Immediately preceding and following a large earthquake, there is a consistently elevated number of events that decays over time following an empirically derived power law relation, i.e. Omori's Law (Omori, 1894; Utsu & Ogata, 1995). As the number of dependent events tends to obscure the background seismicity rate, they strongly influence many statistical analyses of seismicity if they are not effectively removed. Such removal of dependent events is called declustering and is an important part of studies investigating probabilistic earthquake hazards, temporal patterns in seismicity, clustered seismicity models, or earthquake prediction research (van Stiphout et al., 2012). As such, the main purpose of declustering algorithms is to identify either clusters of earthquakes or background events, such that dependent events can be removed from a catalog, yielding a realistic background seismicity rate and preserving the temporal distribution of the independent events.

However, the lack of a 'ground truth' or record of all truly independent events precludes easily determining a best algorithm to use in any given circumstance. Complicated by assumptions on what background seismicity should look like, declustering algorithms all yield highly non-unique solutions. This becomes especially problematic as choice of declustering algorithm is often arbitrary, with little apparent concern given to the formulation of the algorithm of choice and the variability that stems from tunable parameters. A common assumption is that background seismicity (excluding earthquake swarms) can be modeled by a temporally

stationary Poisson process (Gardner & Knopoff, 1974; Luen & Stark, 2012). Such a process models background seismicity as a random process in time, but not necessarily space. While this is a reasonable model, it is not necessarily true in all locations. For example, seasonal forcing from hydrologic loading and natural variations in pore pressure (e.g. Christiansen et al., 2005; Heki, 2003; Johnson et al., 2017b, 2020) as well as artificial variations in pore pressure from wastewater injection (Teng & Baker, 2019; Zhai et al., 2019) introduce variability (both natural and anthropogenic) to stress conditions that entrain the background seismicity in a given region. Additionally, far field triggering of seismicity (e.g. Freed, 2005), and evidence for global synchronization of earthquakes (Bendick & Bilham, 2017; Bendick & Mencin, 2020) both cast doubt on the Poisson process hypothesis for large scale systems.

While the determination of an optimum declustering algorithm has been investigated before, prior studies have been spatially limited to one region, generally those with many high quality observations (Llenos and Michael, 2020; Mizrahi et al., 2021; Teng and Baker, 2019; van Stiphout et al., 2012). To mitigate these issues, we investigate the results of four of the most commonly implemented declustering algorithms: Gardner and Knopoff (1974), Reasenberg (1985), Zhuang (2002), and Zaliapin and Ben-Zion (2013a, 2013b, 2020), employed in four tectonic settings: the Northern Rocky Mountains (excluding Yellowstone), Yellowstone, Southern California, and Japan. In doing this we hope to determine which algorithm is most suitable for declustering purposes in any given tectonic setting, as well as the potential benefits and pitfalls of each algorithm, which can then be used to determine which algorithm is most suitable for a given study.

2. Declustering Algorithms

The Gardner & Knopoff (1974) algorithm uses a windowing method, where a region in space around the earthquake and time after the earthquake is calculated based on the magnitude of the mainshock, as shown in Eq. (1).

$$d = 10^{0.1238*M+0.983}t = \begin{cases} 10^{0.032*M+2.7389}, & \text{if } M \geq 6.5 \\ 10^{0.5409*M-0.547}, & \text{if } M < 6.5 \end{cases} \quad (1)$$

Here, M is the magnitude of a mainshock, while d is distance in kilometers, and t is time in days, from a mainshock that define the window (van Stiphout et al., 2012). All events that fall into this window are then assumed to be aftershocks of the mainshock and are removed from the catalog. While this is a fast and robust method, it is easy to see how this algorithm might cull more events than necessary, as any background events within the calculated window would be assumed to be aftershocks. Additionally, another weakness of this method is that it cannot account for more complex chains of cascading events (i.e. aftershocks of aftershocks), and thus may underestimate the number of aftershocks in certain circumstances. Lastly, while there are no parameters to be estimated by the user in this method, its simple construction makes it relatively simple to modify the window size if there is evidence in the output catalog that indicate poor declustering.

The Reasenberg (1985) algorithm improves upon the Gardner and Knopoff (1974) method in multiple ways. Most importantly, it allows for higher order aftershocks to form clusters of events that are then removed except for the mainshock. While it still uses a windowing method to define local events, this is not an empirically

derived relation, but is instead based off of a simple circular fault model and defined in Eq. (2) (Molchan & Dmitrieva, 1992).

$$\log(d) = 0.4M_0 - 1.943 + k \quad (2)$$

In Eq. (2), d is the distance from an event in kilometers that aftershocks are likely to occur in, M_0 is the moment magnitude of an earthquake, and k is a parameter that is equal to 1 if the event is the largest in a sequence, 0 if it is not. Additionally, this method implements Omori's law for determining the temporal distance between an event and its dependent events without any assumptions on the spatial structure of an aftershock sequence (Molchan & Dmitrieva, 1992). Lastly, Reasenbergs declustering requires the user to estimate and input six different parameters, that include factors used in estimating the temporal windows (τ_{min} and τ_{max}), interactions radii (r_{fact}), the parameters defining lower magnitude threshold of the catalog in question and how it changes after large mainshocks (x_{meff} and x_k), as well as statistical parameters about the accuracy of event times in the catalog (p). While Schorlemmer & Gerstenberger (2007) provide respective ranges for each of the parameters, there is no consensus on optimum parameters, and the defaults are often used.

The method developed by Zhuang et al. (2002) differs from the previous algorithms mentioned in this study in that it is a stochastic, not deterministic method. Assuming a stationary background seismicity rate, this method calculates the probability of any event to be a background event based off the space-time branching epidemic type aftershock sequence model (the ETAS model) established by Ogata (1988, 1998). While this declustering method requires the estimation of eight different parameters, the algorithm uses a maximum likelihood estimation (MLE) to converge on best-fit parameters for an ETAS model given an input catalog, thus eliminating the need for the largely arbitrary parameter estimation present in other declustering algorithms.

Zaliapin et al. (2008) developed a method to decluster earthquakes using a nearest-neighbor approach by leveraging the interevent space-time distance of Baiesi & Paczuski (2004), further developed in subsequent work (Zaliapin & Ben-Zion, 2013a, 2013b; Zaliapin & Ben-Zion, 2020). This algorithm effectively calculates interevent "distance" using Eq. (3).

$$\eta_{ij} = \begin{cases} t_{ij}(r_{ij})^{d_f} 10^{-wm_i}, & t_{ij} > 0 \\ \infty, & t_{ij} \leq 0 \end{cases} \quad (3)$$

Here, η_{ij} is the interevent space-time distance between event i and j , $t_{ij} = t_j - t_i$ is the interevent time in years between, and can be positive or negative depending on which event occurred first, r_{ij} is the distance between earthquakes in kilometers, d_f is the fractal dimension of the earthquake hypocenters, w is a parameter that introduces the exponential weight of the previous event, and m_i is the magnitude of earthquake i . It is important to note that for cluster identification purposes, w is generally equal to the Gutenberg-Richter b -value of the catalog, but for declustering applications of this algorithm $w = 0$ (Zaliapin & Ben-Zion, 2020). Once these interevent space-time distances have been calculated, nearest neighbors are found for each event that occurs before them in time. This yields a bimodal distribution of the nearest neighbor distances, where the larger mode indicates background events either with or without associated aftershocks. In practice, this algorithm only requires the user to estimate and input three parameters (d_f , w , and α_0 where α_0 is the cluster

threshold and is generally taken to be near zero for declustering purposes) two of which (d_f and w) do not strongly influence the output declustered catalog (Zaliapin & Ben-Zion, 2020).

3. Data

To determine the optimal declustering algorithm to use, we test four different declustering algorithms that are commonly used today. All declustering algorithms (Gardner and Knopoff (1974), Reasenber (1985), Zhuang (2002), and Zaliapin and Ben-Zion (2008, 2013a, 2013b, 2020)) are tested on four earthquake catalogs from different tectonic settings: the Northern Rocky Mountains (excluding Yellowstone), the broader Yellowstone region, Southern California, and Japan (Fig. 1).

The northern Rocky Mountains catalog spans 2014 until 2020 and contains hypocenters from Montana, Idaho, and part of Wyoming. It contains multiple tectonic features including the northern portion of the Intermountain Seismic Belt and the Centennial Tectonic Zone (Fig. 1A). The intraplate region is characterized by basin and range style extension and orogenic collapse (Faulds & Varga, 1998; Payne et al., 2012, 2013; Schmeelk et al., 2017), resulting in seismically active normal and strike-slip faults. We use hypocenters from the United State Geological Survey (USGS) catalog (ANSS comcat) (<https://earthquake.usgs.gov/earthquakes/search>) above the magnitude of completeness ($M_c = 1.1$) from the region spanning all of Idaho north of the Snake River Plain and Montana west of longitude -109.5 (Fig. 1A). Lastly, the region containing the Yellowstone Caldera is removed from this catalog in order to avoid any influences on the declustering parameters from magmatic signals. The Yellowstone catalog is sourced directly from the northern Rocky Mountains catalog, running from 2014 to 2020 and containing events that occurred within the broadly defined Yellowstone region (Fig. 1A). The southern California catalog spans 2010 until 2021 and contains all of California south of latitude 35.5 to latitude 31.5 , and west of -114.5 to longitude -121.5 (Fig. 1B), and includes significant portions of the San Andreas Fault, San Jacinto, and Garlock fault systems. Seismicity in the region is characterized by strike-slip and dip-slip faulting. Similar to the northern Rockies, we use hypocenters from the ANSS comcat, limited to only include events $M \geq 2.0$. Lastly, the Japanese catalog runs from 2010 until 2019 and is sourced from the Japanese Meteorologic Agency (JMA) catalog (https://www.data.jma.go.jp/svd/eqev/data/bulletin/index_e.html), including events from both the Japanese, Ryukyu, and Kuril subduction zones. The region this catalog covers is large, from longitude 120 in the east to 155 in the west, and between latitudes 20 and 50 (Fig. 1C). Additionally, this catalog is also limited to only include events $M \geq 4.0$ and at a maximum depth of 800 km.

For each catalog we also select a mainshock for use in analysis of its aftershock sequence. Each mainshock is generally the largest event in a catalog with sufficient data following the event to constrain Omori's law for aftershock decay. The mainshock for the northern Rocky Mountains is the 2017 M_w 5.8 Lincoln, MT event (McMahon et al., 2019). The largest event in the Yellowstone catalog is a M_w 4.8 event that occurred in early 2014, though considering the more swarm-like seismic activity of the region (e.g. Farrell et al., 2009), a typical aftershock sequence is not expected. The largest event in the southern California catalog is the 2010 M_w 7.2 El Mayor Cucapah event (Wei et al., 2011). Lastly, the largest event in the Japanese catalog is the 2010 M_w 9.0 Tohoku event (Ozawa et al., 2011).

4. Methods

In order to test the effectiveness of each declustering algorithm, we run each algorithm for the four different earthquake catalogs. All algorithms are run with default parameters to assess how well each algorithm works under standard conditions on the separate test datasets. Lastly, as Gardner and Knopoff (1974), Reasenber (1985), and Zaliapin and Ben-Zion (2008) all determine which events are likely to be mainshocks or single events, these are taken to consist of the declustered catalog. Zhuang et al. (2002), on the other hand, is a stochastic declustering algorithm and thus generates the probability that any given event is a background event. Here, we take all events with an arbitrarily chosen background probability of at least 75% to make up the declustered catalog.

Cumulative event vs. time plots from the resulting declustered catalogs are then visually inspected and compared to that of the raw catalogs to determine if the declustering algorithms had successfully removed the vast majority of the aftershock sequences. In order to quantify this, we also fit a modified Omori-Utsu decay law (Eq. (4)) to the raw and declustered catalogs for a period of one year after a given mainshock in the catalog.

$$N = \frac{k}{(t + c)^p} \quad (4)$$

N is the number of events, t is the time since a mainshock, k and c are empirically derived constants, and p represents the logarithmic decay rate. For most aftershock sequences $p \approx 1$ (Utsu & Ogata, 1995), though there does exist some variation. Due to the nature of declustering catalogs, we expect the fit to be exceptionally poor if the algorithm worked correctly and yield a $p \approx 0$, representing a catalog at or near background seismic levels. In contrast, being able to fit an Omori-Utsu decay law indicates that the catalog was not effectively declustered. It is important to note that in fitting Omori's law to an aftershock sequence, we do not use a subset of the catalog that have been classified as aftershocks nor events that occurred within a specified distance from a mainshock. Instead, we use the entire catalog in an effort to mitigate the advantage that windowing procedures would have due to the total elimination of all events near a mainshock. As a result, our reported p values are skewed by the presence of significant background seismicity far from the mainshock.

Additionally, we also compare the Gutenberg-Richter (GR) parameters (Gutenberg & Richter, 1956) of the raw catalogs to that of the declustered catalogs to assess which catalogs best maintained the original frequency-magnitude relation, Eq. (5).

$$N = 10^{a-bM} \quad (5)$$

Here, N is the total number of events expected of a given magnitude M in a catalog, while a and b are empirically derived values that represent the total seismicity rate of a given region and the ratio of large events to small events, respectively. We assume that declustering algorithms that produce catalogs with similar GR parameters as the respective raw catalogs are better representations of actual seismicity in any given region, though as declustering removes a significant number of events we do expect these values to change.

Furthermore, we also calculate the coefficient of variation (C_v) of interevent times for each catalog (Eq. 6).

$$C_v = \frac{T_{SD}}{\bar{T}}(6)$$

Defined as the ratio of the standard deviation of interevent times of a catalog (T_{SD}) and the mean of the interevent times ($\bar{T}$), the coefficient of variation is useful in determining if an earthquake catalog is Poissonian in time. When a catalog is near Poissonian, C_v will be ~ 1 , a $C_v > 1$ indicates that the catalog is increasingly fractal, while $C_v < 1$ indicates that a catalog is increasingly periodic (Kagan & Jackson, 1991).

Finally, we also compare the proportion of events removed from all declustered catalogs. Working under the assumption that more data are better, and more events are likely to represent the actual seismic productivity of a given region provided that aftershock sequences are successfully removed. In summary, we prefer catalogs that remove the fewest events, destroy any semblance of the Omori-Utsu law, and maintain the GR parameters. While we also calculate a measure of how Poissonian a catalog is, we do not hold this metric to essential in determining the most effective declustering algorithm for any given region; rather the coefficient of variation might be of distinct research interest if it characterizes differences in temporal event organization from region to region.

5. Results

Our results are summarized in the cumulative event vs. time plots (Figure 2) and in Table 1. In all instances, the Zaliapin and Ben Zion method effectively removes aftershocks, while simultaneously allowing for the most events to exist in the resulting catalog. Reasenbergs declustering tends to remove the fewest events, but there exist easily identifiable deviations from a linear trend in the background seismicity (Figure 2) as well as a still intact fit to Omori's law (that is a decay parameter ≥ 0.2) in two of the four locations tested (Table 1). This indicates that the algorithm, when using default parameters, does not fully remove aftershock sequences. In some cases, the best fit Omori p value is negative indicating that the number of earthquakes is increasing with time and the declustering algorithm may have removed more events near a mainshock than ideal, though the magnitude of this is small ($p > -0.25$) in all cases where p is negative. In comparison, while both Gardner and Knopoff and Zhuang-ETAS algorithms effectively remove aftershock sequences in all catalogs, the resulting declustered catalogs have

Table 1 Summary of Results including calculated Gutenberg-Richter a and b values, as well as the change in a and b ($\Delta a, \Delta b$) in comparison to that of the raw catalog. We also report the calculated omori- p value for the entire catalog following the largest mainshock, as well as the coefficient of variation.

Catalog	Method	a (GR)	Δa (GR)	b (GR)	Δb (GR)	Omorip	Cv	Total Events	% removed
Northern Rockies, 2014-2020	Raw	0.131	0.000	0.808	0.000	0.456	1.340	6524	0.0
	Zaliapin	0.006	0.125	0.742	0.066	0.005	1.030	3668	43.8
	Zhuang	0.226	-0.095	0.871	-0.063	-0.002	1.104	1576	75.8
	Reasenber	0.084	0.047	0.794	0.014	0.231	1.146	5222	20.0
	GK	-0.023	0.154	0.742	0.066	-0.062	1.025	2891	55.7
Yellowstone, 2014-2020	Raw	0.285	0.000	0.916	0.000	0.283	2.233	4074	0.0
	Zaliapin	-0.053	0.338	0.724	0.192	0.045	1.005	1150	71.8
	Zhuang	0.432	-0.147	1.039	-0.123	0.037	4.830	461	88.7
	Reasenber	0.587	-0.302	1.104	-0.189	0.157	1.238	1659	59.3
	GK	-0.117	0.402	0.682	0.234	-0.237	1.025	804	80.3
Southern California. 2010-2021	Raw	1.028	0.000	0.901	0.000	0.812	1.909	21448	0.0
	Zaliapin	0.718	0.310	0.798	0.103	-0.204	0.995	5817	72.9
	Zhuang	1.472	-0.444	1.058	-0.157	0.125	0.987	3261	84.8
	Reasenber	1.589	-0.561	1.091	-0.190	0.812	1.531	14816	30.9
	GK	0.545	0.483	0.705	0.196	0.545	0.979	2892	86.5
Japan, 2010-2018	Raw	3.223	0.000	0.968	0.000	0.865	1.488	15894	0.0
	Zaliapin	2.882	0.341	0.895	0.073	-0.097	0.986	6776	57.4
	Zhuang	3.162	0.062	0.972	-0.003	-0.078	0.990	4720	70.3
	Reasenber	3.541	-0.318	1.046	-0.077	0.026	0.384	2855	82.0
	GK	2.152	1.071	0.738	0.230	0.138	1.018	3018	81.0

far fewer events than the catalog declustered using the Zaliapin and Ben-Zion method.

Additionally, we find that the magnitude of changes in the best-fit post-declustering GR parameters are generally not significantly different between declustering mechanisms (Table 1). Lastly, the C_v indicates that both the Zaliapin & Ben-Zion and Gardner & Knopoff methods yield relatively Poissonian catalogs in all test cases. Meanwhile, Reasenber yields non-Poissonian California and Japan catalogs, while Zhuang-ETAS produces a very non-Poissonian Yellowstone catalog (Table 1).

Interestingly, in all catalogs except the Japanese catalog, a ranking of all methods in terms of events retained is stable. Reasenber declustering yields the most events in the output catalogs although the resulting catalogs still maintain artifacts of aftershock sequences (Figure 2). This is followed by Zaliapin and Ben-Zion, which removes easily identifiable aftershock sequences, yet removes the fewest events. Gardner and Knopoff

and the Zhuang-ETAS method follow with similarly stable rates of background seismic productivity yet significantly fewer events than the other methods. Notably, there is also a slight increase in the background seismicity rate produced by the Zaliapin and Ben-Zion method (greater than other methods) in the declustered Japanese and Southern California catalogs following the 2011 Tohoku and the 2010 El Mayor Cucapah events respectively (Figure 3).

6. Discussion

Of the four declustering algorithms tested in the four different test regions, we find the method of Zaliapin and Ben-Zion (2008) to be the most effective removing aftershock sequences, while simultaneously allowing for the most events to remain in the output catalog. However, this method does have some interesting features that may alter its effectiveness in a given region. As noted by Teng & Baker (2019), the Zaliapin and Ben-Zion algorithm is able to account for temporally varying seismic productivity in regions like Oklahoma, where induced seismicity dominates. Similarly, we also see an increase in seismic productivity following the 2011 Tohoku earthquake in the Japanese catalog, as well as similar elevated seismic productivity following the 2010 El Mayor Cucapah earthquake in the Southern California catalog after a decrease in seismic productivity immediately following the event (Fig. 3). It is important to note that these increases in seismic productivity are small, no more than $\sim 2\%$ of the best-fit stable seismicity rate (Fig. 3). Regardless, this indicates that the Zaliapin and Ben-Zion (2008) method may prove insufficient to effectively remove changes in background seismicity following large events where Omori-Utsu would predict elevated seismicity for several years. However, following the results of Teng & Baker (2019), utilizing an algorithm like Zaliapin and Ben-Zion that can account for changes in background seismic productivity over multiple years is likely more accurate for the purposes of hazard analysis.

It is also important to note that immediately surrounding large magnitude events in all catalogs except Yellowstone there is a decrease in seismic productivity when declustered using the Zaliapin and Ben-Zion method (Fig. 3). This, combined with the negative best-fit Omori- p values for the California and Japanese catalogs (Table 1), indicate that Zaliapin and Ben-Zion declustering removes a proportionally greater amount of events surrounding a mainshock than the other methods.

Reasenbergs declustering, despite the large number of events preserved in the output catalogs, does not appear to be a particularly useful method, provided that it has a tendency to leave aftershock sequences intact in the output catalogs (Table 1, Fig. 2). Additionally, testing of various parameters within the limits proposed by Schorlemmer & Gerstenberger, (2007) indicates that even with the optimal tuning of parameters, Reasenbergs declustering tends to insufficiently decluster the tested earthquake catalogs. Of the catalogs tested in this study, only the Japanese catalog was reasonably declustered. This is potentially because in the generating the temporal interaction zones of an earthquake, Reasenbergs assumes an Omori p value of 1, thus it will work best in regions where such a value is close to reality, like Japan (e.g. Enescu et al., 2009; Table 1). Additionally, the influence of the large 2011 Tohoku event, and the relatively high lower magnitude limit of our test catalog limit ensures that the catalog is complete and branching structures within the catalog can be fully traced to effectively remove an entire aftershock sequence.

Though they tend to remove more events than the other two methods, both Gardner and Knopoff and Zhuang-ETAS effectively remove aftershock sequences in all test areas (Fig. 2), and are thus also useful for declustering, especially in regions where a small catalog is not a limitation. That said, the very nature of the Gardner and Knopoff declustering, being an algorithm that windows in space and time, removes all events near a large earthquake and thus is liable to remove events that are actually background seismicity. This is also reflected in the change in the GR-b value where output catalogs from Gardner and Knopoff declustering consistently yield the largest magnitude of Δ GR-b (Table 1), indicating a consistent change in the statistical distribution of events that increases the expected occurrence of large magnitude events. While Mizrahi et al. (2021) found that the Zhuang-ETAS method is best in terms of preserving the GR-b value in the declustered catalogs, we only find a significant difference in the magnitude of the change in the GR-b value in one region, Japan (Table 1). Additionally, the loss of data from this method makes it particularly unsuitable in regions that are not well instrumented, especially when analyzing an earthquake catalog for anything other than a background seismicity rate. Furthermore, the poor performance of the Zhuang-ETAS method in reproducing a relatively poissonian catalog in regions where the general seismic pattern is more swarm-like (ie Yellowstone, Table 1) indicates that it may not be widely applicable. While declustering based off of regionally optimized ETAS models (e.g. Llenos and Michael, 2019) may rectify these issues, this is not investigated here. Another interesting feature of the declustered catalogs produced using the Zhuang-ETAS algorithm is that the largest events in the catalog often have very small background probabilities and thus are unlikely to be included in the declustered catalog. For some purposes of declustering, this is not a large problem because one only needs an estimate of background seismicity, though a declustered catalog that does not include the largest events will likely skew probabilistic estimates on the potential for large events. For both these reasons, we do not recommend this method in regions where large losses in data are likely to change the statistical characteristics of the catalog or when one needs to estimate the probability of large magnitude events.

7. Summary And Conclusions

While all declustering algorithms reduce the number of aftershocks in a catalog and can be used successfully assuming proper parameter estimation, some tend to perform better than others. The Gardner and Knopoff windowing algorithm is able to remove aftershock sequences in all test regions, though it tends to remove more events than other algorithms tested, likely due to the imprecise nature of windowing methods. Reasenberg declustering is unable to effectively remove remnants of aftershock sequences in three of the four test regions (Fig. 2), thus we do not recommend its use in most circumstances. The Zhuang-ETAS algorithm is successful in removing aftershock sequences in all tested locations, though it tends to remove more events than other algorithms tested. The Zaliapin and Ben Zion algorithm works well to remove aftershock sequences without removing many events. However, subtly elevated background seismicity exists following large events in the two tested catalogs with mainshocks of $M \geq 7$.

Declarations

All authors have consented to submission and publication of this manuscript. The authors declare no conflicts of interest relevant to this research.

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Author's Contributions

Mason Perry and Rebecca Bendick wrote the manuscript text. Mason Perry conducted statistical analysis and prepared the figures. All authors have reviewed the manuscript.

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Figures

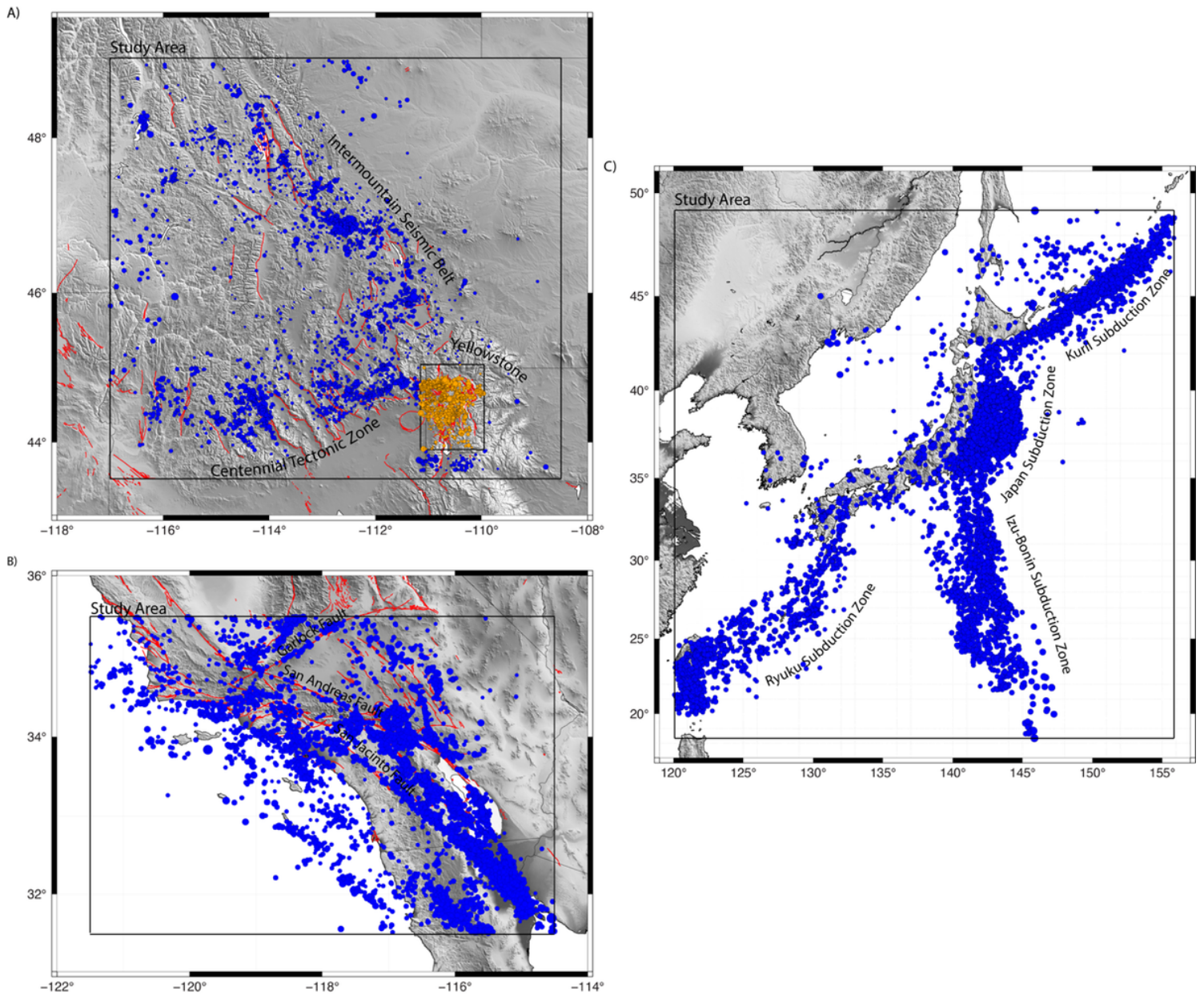


Figure 1

Overview of the four study regions (black boxes), all earthquakes in raw catalogs are plotted and sized by magnitude. Mapped faults in the northern Rockies and California are plotted in red. Major tectonic features are labeled. 1a) Map of the northern Rockies with raw catalogs from the northern Rockies (blue) and Yellowstone (orange). 1b) Map of Southern California with the raw earthquake catalog plotted. 1c) Map of Japan plotted with the Japanese raw earthquake catalog plotted

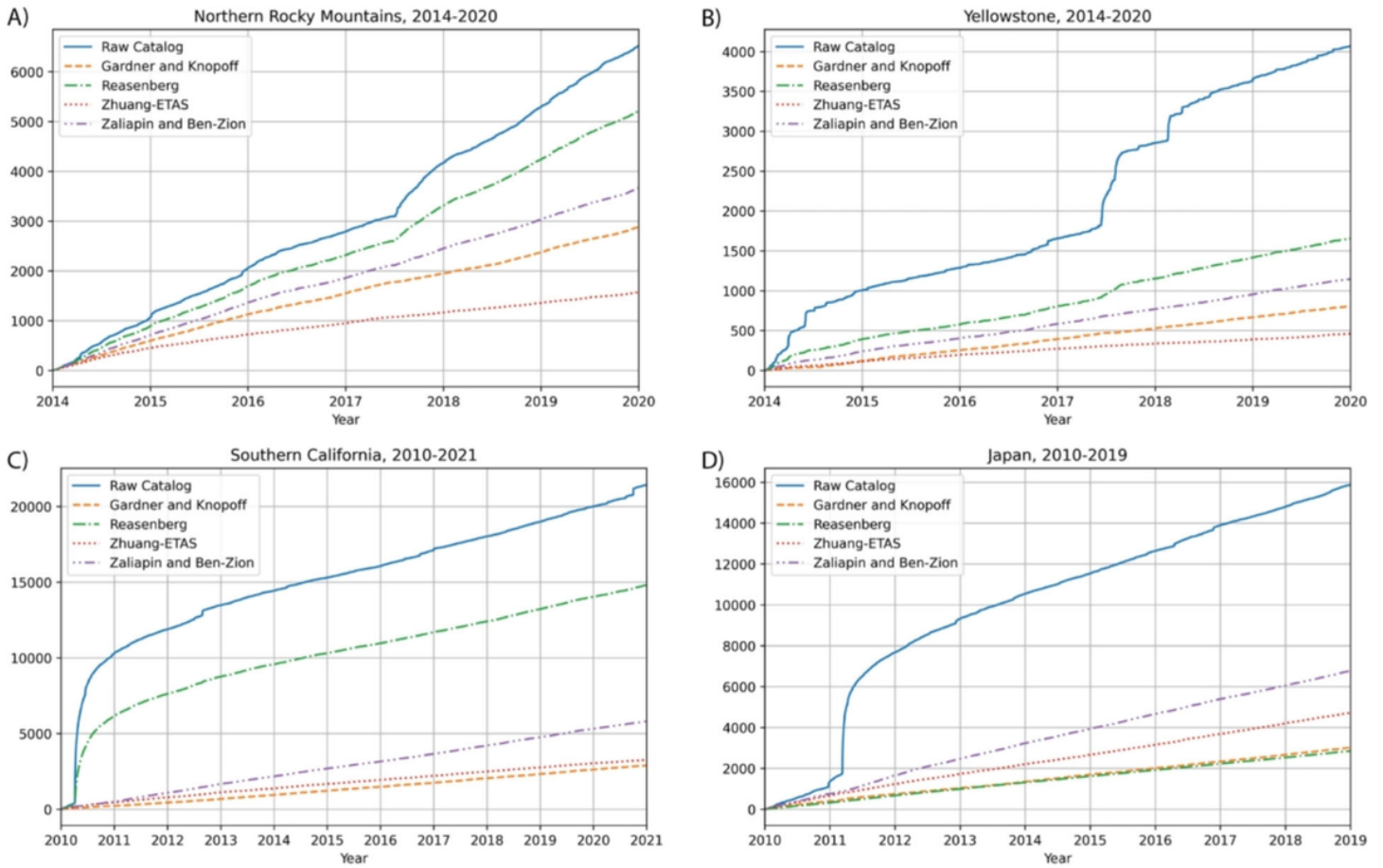


Figure 2

Plots of cumulative events over time comparing raw and declustered catalogs. In all plots, the raw catalog exhibits significant seismicity rate variations. 2a) The Northern Rocky Mountains, 2b) Yellowstone, 2c) Southern California, and 2d) Japan. In all plots (except for the Japanese catalog), Reasenberg declustering does not fully remove an aftershock sequence, while all other declustering algorithms do

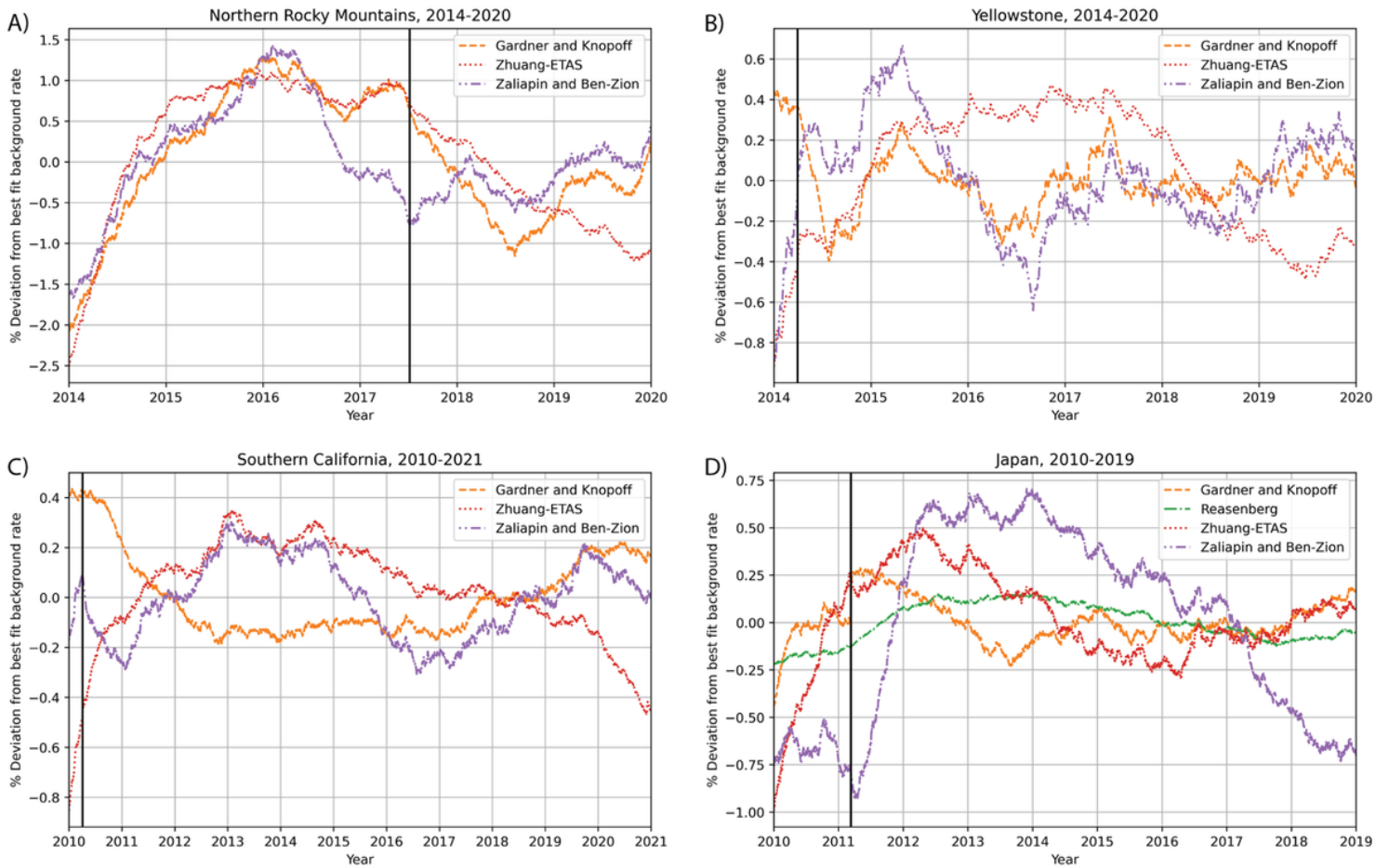


Figure 3

Plots of percent deviation from best-fit stable background rate. On each plot, the time of the largest earthquake (i.e. mainshock), is marked with a black vertical line. To avoid saturating the plots, plots of raw catalogs are not included in any of the plots. For the same reasons, Reasenberg is only included in the Japan plot. 3a) Northern Rocky Mountains, 3b) Yellowstone, 3c) Southern California, and 3d) Japan